

MDA: A Mail Delivery Agent or Message Delivery Agent is a piece of software that is responsible for the delivery of the messages or mail to a recipient's mailbox.

What is to be installed?

To have fully operational email capabilities, we'll need to install the following:

Postfix: A MTA and the most popular alternative to Sendmail that was released by Wietse Venema in December 1998

Dovecot: A mail server suite that includes a MDA, an IMAP and POP3 server. It was released in 2002 by Timo Sirainen.

SpamAssassin: An email spam filtering software originally written by Justin Mason, and which is part of the Apache Foundation.

SquirrelMail: A Web mail interface originally written by Nathan and Luke Ehresman.

Setting up the Virtual Private Server (VPS)

We'll need a VPS and the smaller the better. A 20 GB HDD and 512 MB of memory should suffice. Digital Ocean, Rackspace and Tata Insta Compute have such offerings, though there must be other providers with similar options. Spin an *instance* with the bare minimum configuration using CentOS 6.x. If you prefer Ubuntu, you can spin an Ubuntu *instance*, but the rest of the article assumes that you have CentOS installed on your VPS.

Most Linux distributions have Sendmail running by default. Check if Sendmail is running on your VPS and remove it. We'll install Postfix to do its job.

```
ps aux | grep sendmail
yum remove sendmail
```

Set up the fully configured domain name as the host name by executing the following command:

```
echo "HOSTNAME=sastratechnologies.in" >> /etc/sysconfig/network
```

Alternatively, open the network file using an editor like Nano or Vi, and type in the host name, i.e., replace *sastratechnologies.in* with your domain name.

```
nano /etc/sysconfig/network
HOSTNAME=sastratechnologies.in
```

Next, open the *hosts* file using Nano or Vi, and add a host entry for your domain. In the example below, replace the IP address and the domain with your IP address and domain:

```
nano /etc/hosts file
146.185.133.41 sastratechnologies.in
```

Change the time zone of your VPS by executing this command:

```
ln -sf /usr/share/zoneinfo/Asia/Kolkata /etc/localtime
```

Next, set up your iptables to allow incoming and outgoing connections on ports 25, 110, 143, 465, 587, 993 and 995. You can put these commands in a file and execute them:

```
iptables -F
iptables -A INPUT -p tcp --tcp-flags ALL NONE -j DROP
ip tables -A INPUT -p tcp! --syn -m state --state NEW -j DROP
ip tables -A INPUT -p tcp --tcp-flags ALL ALL -j DROP
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -p tcp -m tcp --dport 25 -j ACCEPT
iptables -A INPUT -p tcp -m tcp --dport 110 -j ACCEPT
iptables -A INPUT -p tcp -m tcp --dport 143 -j ACCEPT
iptables -A INPUT -p tcp -m tcp --dport 465 -j ACCEPT
iptables -A INPUT -p tcp -m tcp --dport 487 -j ACCEPT
iptables -A INPUT -p tcp -m tcp --dport 993 -j ACCEPT
iptables -A INPUT -p tcp -m tcp --dport 995 -j ACCEPT
iptables -I INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -P OUTPUT ACCEPT
iptables -P INPUT DROP
iptables -L -n
iptables-save | sudo tee /etc/sysconfig/iptables
service iptables restart
```

We have completed setting up the VPS. We should now install the postfix MTA. Execute this command from the terminal:

```
yum -y install postfix
```

You should see some messages on the screen while it is getting installed.

Installing SMTP authentication and creating certificates

Having installed Postfix, let's now install the SMTP AUTH packages, which provide a SSL channel for your SMTP server. Install the packages by executing the following command from the terminal:

```
yum -y install cyrus-sasl cyrus-sasl-devel cyrus-sasl-gssapi
cyrus-sasl-md5 cyrus-sasl-plain
```

Once the SMTP AUTH packages are installed, create the directories required for storing the *ssl* certificates:

```
mkdir /etc/postfix/ssl
cd /etc/postfix/ssl/
```

Now generate a private key. You will be prompted to enter a pass phrase. Provide a password, write it down and keep it in a safe place.